Milestone 2 – Technical Checkpoint

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Milestone 2 – Technical Checkpoint

Project Title

Data-Driven Power System Upgrade Planning for Cellular Network Sites Using Traffic and Hardware Configuration

1. Introduction

This project aims to develop a data-driven decision-support system for power system upgrade planning in cellular network sites, leveraging traffic telemetry, hardware configuration, and power consumption data. As defined in Milestone 1, the core objective is to predict site-level power requirements, detect anomalies, and support infrastructure planning decisions.

Milestone 2 focuses on demonstrating technical progress, including data preprocessing, exploratory analysis, initial modeling, and system design.

2. Data Acquisition Challenges and Mitigation Operational Data Lake

2.1. Original data plan

The initial proposal relied on extracting,

- 15-minute interval traffic data over 6 months
- Historical power consumption
- Hardware configuration data

from a Sri Lankan mobile network operator's internal data lake.

2.2. Issue encountered

Due to,

- Strict data privacy policies
- Complex access approval procedures
- Organizational constraints

it was not possible to obtain the required dataset within the expected timeline. This aligns with a key risk identified in Milestone 1: "Confidentiality constraints limiting public data release".

2.3. Mitigation strategy

To ensure continued progress, an alternative partial set of date was obtained from Cable & Wireless Seychelles (CWS). The dataset includes,

- Multi-RAT data (LTE and 5G NR) – Received except for 2G and 3G
- Site-level and cell-level information – Received in total
- Geographic (GPS) data – Received in total
- Hardware configuration attributes – Not yet received
- Details of upcoming upgrades and installations – Not yet received

2.4. Current dataset scope

Feature	Current Status
Time resolution	15 minutes
Duration	7 days
Technologies	LTE + NR (expecting to add 2G and 3G
Granularity	Site + Cell level
Data type	Time-series + tabular
Data Source	C&WS Data lake

Table 2-1: Current Dataset Scope

2.5. Limitation

While the dataset structure matches the original requirement, the temporal depth is limited (1 week instead of 6 months). This impacts,

- Long-term forecasting accuracy
- Seasonal pattern detection

However, the dataset is sufficient to,

- Build and validate the pipeline architecture
- Demonstrate modeling capability
- Perform short-term analysis

3. Data Preprocessing

3.1. Data Integration

Multiple datasets were integrated:

- NR traffic and power data
- Hardware configuration (yet to)
- Site location (GPS)
- TE traffic and power data

Key identifier: GNodeB_ID used as the unique site key

3.2. Data Cleaning

Steps performed:

- Handling missing values
- Ensuring consistent timestamps
- Aligning 15-minute intervals
- Removing duplicates

3.3. Feature Engineering

Derived features include,

- Daily averages and peak values
- Traffic-to-power relationships
- Site density proxy (based on GPS)

4. Exploratory Data Analysis (EDA)

4.1. Traffic Patterns

Observed patterns,

- Dual-peak behavior (midday and evening)
- Lower usage during night hours
- Variation across sites (urban vs rural)

4.2. Power Consumption Trends

- Strong correlation with traffic load
- Higher consumption in dense areas
- Variation based on hardware configuration

4.3. Cross-Technology Analysis

- NR shows higher throughput than LTE
- LTE traffic capped 80 Mbps
- NR traffic scaled up to 1300 Mbps

4.4. Site Variability

- Urban sites
    - Higher peaks, Greater variability
- Rural sites
    - Lower, stable traffic

4.5. Average Traffic Profile (NR vs LTE)

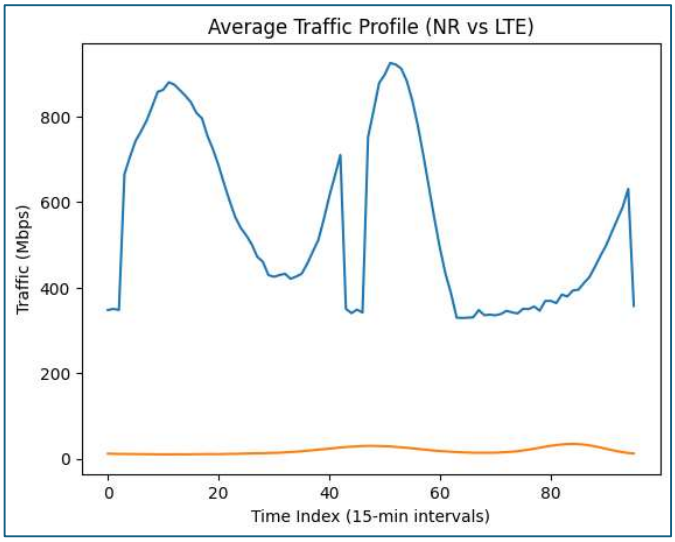

Figure 4-1: Average Traffic Profile (NR vs LTE)

4.6. NR Traffic Distribution

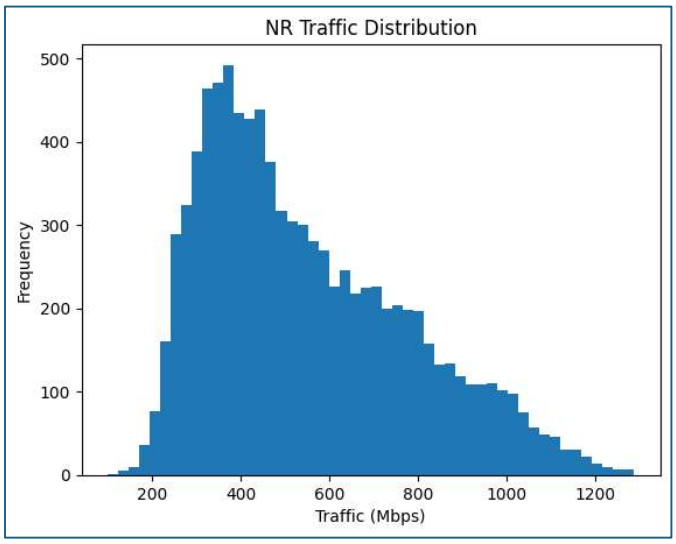

Figure 4-2: NR Traffic Distribution

4.7. NR Traffic vs Power

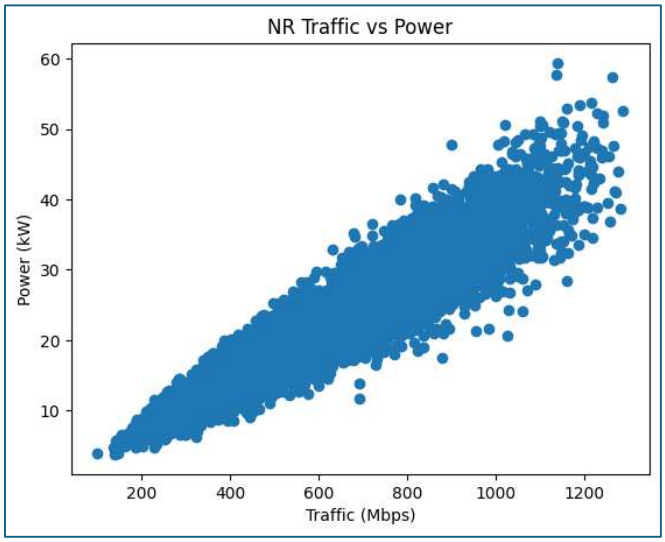

Figure 4-3: NR Traffic vs Power

4.8. LTE Traffic vs Power

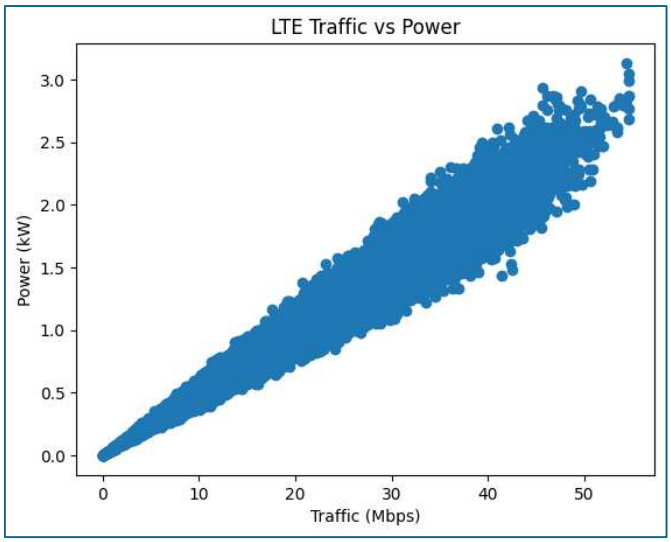

Figure 4-4: LTE Traffic Vs Power

4.9. Traffic Distribution of an NR Cell over time

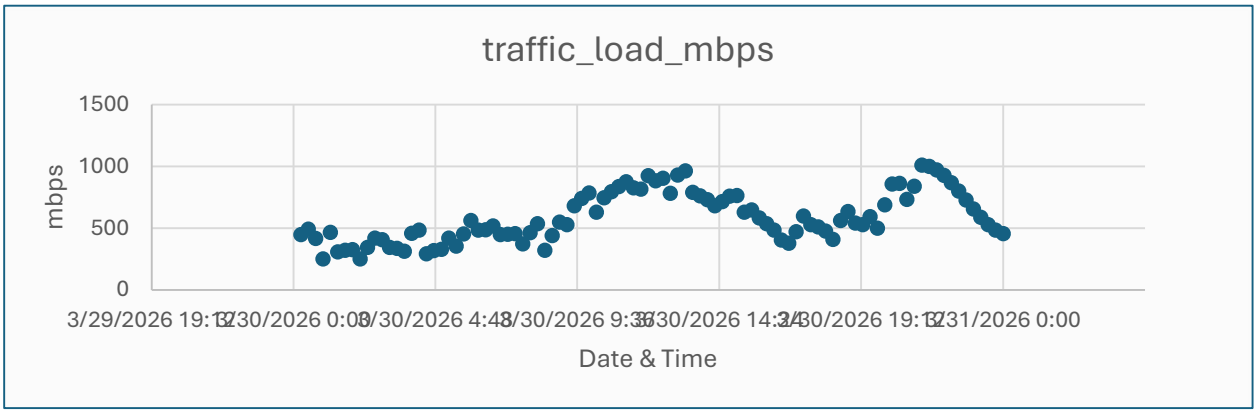

Figure 4-5: LTE Traffic Vs Power

4.10. RF Power Distribution of an NR Cell over time

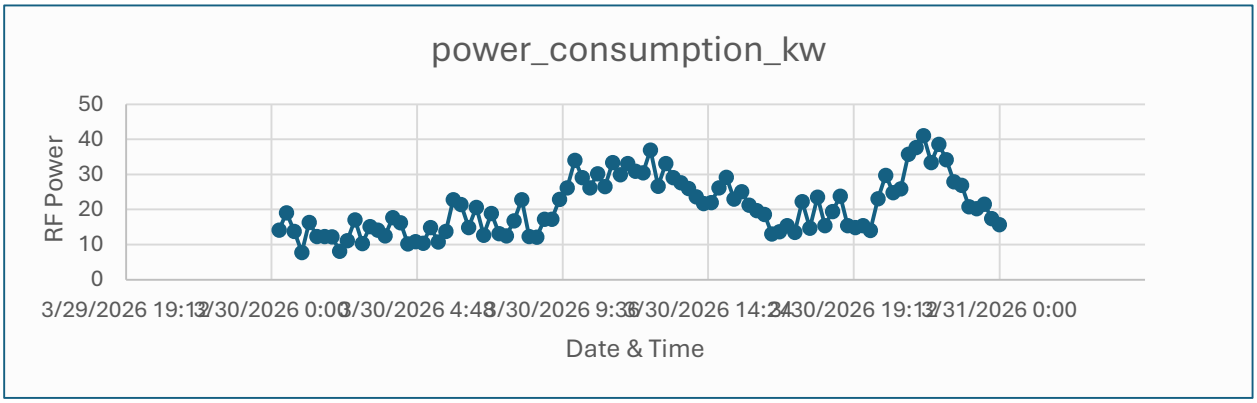

Figure 4-6: NR Traffic Vs Power

5. Location and distribution of sites

Figure 5-1: Location and distribution of sites

6. Baseline and Improved Methods

6.1. Baseline Methods

- Linear regression for power prediction
- Statistical thresholding for anomaly detection

6.2. Improved Methods (In Progress)

- Random Forest regression
- Gradient boosting models
- Isolation Forest for anomaly detection

6.3. Current Status

Due to limited historical depth,

- Models are prototyped but not fully validated
- Performance metrics are preliminary

7. System Architecture

The system follows an end-to-end pipeline,
Data Sources → Preprocessing → Feature Engineering → Modeling → Output Dashboard
Components:

- Data ingestion layer
- Processing and feature extraction
- ML modeling layer
- Output generation

8. Current progress summary

Component	Status
Data preprocessing	10%
Data integration	10%
EDA	50%
Synthetic data generation	0%
Baseline model	0%
Advanced models	0%
System architecture	0%

Table 8-1: Current Progress Summary

9. Gap Analysis

9.1. What is incomplete

- Long-term (6-month) data
- Fully validated forecasting models
- Robust anomaly detection evaluation

9.2. What is achieved

- Data pipeline
- Multi-RAT dataset for 5G (NR) and 4G (LTE)

10. Plan for completion (next 3 weeks)

10.1. Week 1

- Obtain extended dataset or simulate longer history
- Finalize forecasting models

10.2. Week 2

- Complete anomaly detection models
- Perform evaluation (RMSE, MAE, etc.)

10.3. Week 3

- Build final outputs:
  - Site ranking
  - Upgrade recommendations
- Prepare final report

11. Conclusion

Despite challenges in accessing the originally planned dataset, significant technical progress has been achieved. A realistic dataset has been constructed, and the core system pipeline has been implemented. The remaining work primarily involves:

- Extending temporal data depth
- Finalizing and validating models

These are achievable within the remaining project timeline.